

Python for Data Science - 2305CS303

Lab - 4

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1) Character Category Counter

Problem: Count uppercase, lowercase, digits, and special characters in a string.

Sample Inputs & Outputs:

- Input: Python3@Lab → Uppercase: 2, Lowercase: 7, Digits: 1, Special: 1
- Input: HELLO123 → Uppercase: 5, Lowercase: 0, Digits: 3, Special: 0
- Input: hi@#9 → Uppercase: 0, Lowercase: 2, Digits: 1, Special: 2

```
In [6]: i="Python3@Lab"
i2="HELLO123"
i3="HI@#9"
u=0
l=0
d=0
s=0
for ch in i3:
    if ch.isupper():
        u+=1
    elif ch.islower():
        l+=1
    elif ch.isdigit():
        d+=1
    else:
        s+=1
print("count uppercase: ",u," count lowercase: ",l," digits: ",d," special: ",s)

count uppercase:  2  count lowercase:  0  digits:  1  special:  2
```

2) Reverse Each Word

- Input: Python is Easy → nohtyP si ysaE

- Input: Hello World → olleH dlrow
- Input: Learn Python Fast → nraeL nohtyP tsaF

```
In [25]: n=input("enter string: ").split()
for ch in range(len(n)):
    n[ch]=n[ch][-1::-1]
n=" ".join(n)
n
```

enter string: python is easy

Out[25]: 'nohtyp si ysae'

3) Case Pattern Identifier

- PYTHON → Uppercase
- python → Lowercase
- Python Programming → Title Case

```
In [27]: n=input("enter str: ")
if n.isupper():
    print("Uppercase")
elif n.islower():
    print("lowercase")
elif n.istitle():
    print("title case")
```

enter str: PYTHON

Uppercase

4) First and Last Occurrence Finder

- programming , g → First:3 Last:10
- banana , a → First:1 Last:5
- hello , z → Not found

```
In [32]: n=input("enter str: ")
n1=input("enter char: ")
f=n.find(n1)
l=n.rfind(n1)
if f==-1 and l==-1:
    print("Not found")
else:
    print(f,l)
```

enter str: hello

enter char: z

Not found

5) Word Frequency Counter (ignore the case)

- Python is easy and Python is powerful → python: 2
- Java is popular but java is verbose → java: 2
- C is fast → c:1

```
In [34]: n=input("enter str: ").lower()
n1=input("enter word: ").lower()
count=n.count(n1)
print(count)
```

```
enter str: Python is easy and Python is powerful
enter word: python
2
```

6) Remove Extra Spaces

- Python is fun → Python is fun
- Hello World → Hello World
- NoExtraSpace → NoExtraSpace

```
In [40]: n=input("enter str: ")
n=" ".join(n.split())
n
```

```
enter str: Python is fun
```

```
Out[40]: 'Python is fun'
```

7) Prefix and Suffix Removal

- unhappy.txt → happy
- pretest.py → test
- unwanted.doc → wanted

```
In [43]: n=input("enter str: ")
pre=input("enter prefix: ")
suf=input("enter suffix: ")
n=n.removeprefix(pre)
n=n.removesuffix(suf)
print(n)
```

```
enter str: unhappy.txt
enter prefix: un
enter suffix: .txt
happy
```

8) Replace Vowels

- Programming → Pr*gr*mm*ng
- Education → *d*c*t**n
- Sky → Sky

```
In [60]: n=input("enter str: ")
n1="aeiou"
n2=""
for ch in n:
    if ch in n1:
        n2+="*"
    else:
```

```
n2+=ch
n2
```

enter str: Programming

Out[60]: 'Pr*gr*mm*ng'

9) String Compression

- aaabbccccc → a3b2c4
- xxxyyyz → x3y3z1
- abcd → a1b1c1d1

```
In [1]: s = input("Enter string: ")

v = []
r = ""

for i in s:
    if i not in v:
        r += i + str(s.count(i))
        v.append(i)

print(r)
```

a5b2c2

10) Toggle Case (without using str.swapcase())

- PyThOn → pYtHoN
- HELlo → heLLo
- Python → pYTHON

```
In [63]: n=input("enter str: ")
n2=""
for ch in n:
    if ch.isupper():
        n2+=ch.lower()
    elif ch.islower():
        n2+=ch.upper()

n2
```

enter str: Python

Out[63]: 'pYTHON'

11) Username Validation

valid username rules:

1. only alphabets and digits
2. does not start with a digit
3. length ≥ 6

- User123 → Valid
- 123User → Invalid

- Us@12 → Invalid

```
In [81]: n=input("enter str: ")
if n.isalnum():
    if n[0].isdigit():
        print("Invalid")
    else:
        if len(n)>=6:
            print("Valid")
        else:
            print("Invalid")
else:
    print("Invalid")
```

enter str: User123
Valid

12) Palindrome Checker (ignore the case)

- Madam → Palindrome
- Level → Palindrome
- Python → Not Palindrome

```
In [87]: n=input("enter string: ")
n1=n[::-1]
if n==n1:
    print("Palindrome")
else:
    print("not Palindrome")
```

enter string: madam
Palindrome

13) Longest Word Finder

- Python programming is interesting → programming
- I love coding → coding
- Data structures and algorithms → structures

```
In [101]: n=input("enter string: ").split()
max=[]
max2=""
for ch in range(len(n)):
    if len(n[ch])>=len(max2):
        max=ch
        max2=n[ch]
n[max]
```

enter string: Data structures and algorithms

Out[101]: 'algorithms'

14) Case Conversion

- python programming
 - Upper: PYTHON PROGRAMMING

- Lower: python programming
- Title: Python Programming
- Capitalize: Python programming

In [102...

```
n=input("enter str: ")
print(n.upper())
print(n.lower())
print(n.title())
print(n.capitalize())
```

```
enter str: python programming
PYTHON PROGRAMMING
python programming
Python Programming
Python programming
```

15) Custom Split

- apple,banana,grape
- 10|20|30
- A-B-C-D

In [107...

```
n=input("enter str: ")
n1=input("enter split char: ")
```

```
enter str: fdsf
enter split char: fas
```

In []:

In []:

16) Text Formatter

Problem Definition Accept a paragraph and remove extra spaces, convert to Title Case and count words.

Sample

Input:

python programming is interesting

Output:

Formatted Text : Python Programming Is Interesting

Total Words : 4

```
In [2]: s = input("Enter paragraph: ")
```

```
s = " ".join(s.split())
```

```
s = s.title()
```

```
w = len(s.split())
```

```
print("Formatted Text :", s)
```

```
print("Total Words :", w)
```

Formatted Text : Python Is Easy To Learn

Total Words : 5

17) Email Validator

Problem Definition

Validate an email using string methods. A valid email:

- contains exactly one '@'
- contains at least one '.'
- does not start or end with '@' or '.'
- contains no spaces.

Sample

Input: student@gmail.com

Output: Valid Email

```
In [4]: e = input("Enter email: ")
```

```
if e.count("@") == 1 and "." in e and not e.startswith(("@", ".")) and not e.endswith(
```

```
else:
```

```
print("Invalid Email")
```

Invalid Email

18) Username Generator

Problem Definition

Generate username from full name and birth year Generate a username by:

- removing extra spaces,
- converting the name to lowercase,
- replacing spaces with underscores,
- appending the last two digits of the birth year.

Sample

Input:

Name : Rahul Kumar Sharma

Birth Year : 2004

Output:

Username : rahul_kumar_sharma04

```
In [5]: n = input("Name : ")
y = input("Birth Year : ")

n = "_".join(n.split()).lower()

u = n + y[-2:]

print("Username :", u)
```

Username : rak_logo02

19) Sentence Statistics

Problem Definition

Display number of words,vowels,consonants,digits,special chars.

Sample

Input:

Python3 is Awesome!

Output:

Words : 3

Vowels : 6

Consonants : 9

Digits : 1

Special Characters : 1

```
In [6]: s = input("Enter text: ")

w = len(s.split())
v = 0
c = 0
d = 0
sp = 0
```

```

for i in s:
    if i.lower() in "aeiou":
        v += 1
    elif i.isalpha():
        c += 1
    elif i.isdigit():
        d += 1
    elif i != " ":
        sp += 1

print("Words :", w)
print("Vowels :", v)
print("Consonants :", c)
print("Digits :", d)
print("Special Characters :", sp)

```

Words : 3
 Vowels : 6
 Consonants : 10
 Digits : 0
 Special Characters : 0

20) Word Replacement Utility

Problem Definition

Accept a sentence, a word to search, and a replacement word. Replace only if the word exists; otherwise display an appropriate message.

Sample

Input:

Sentence : Python is easy to learn.

Search : easy

Replace : powerful

Output:

Updated Sentence : Python is powerful to learn.

```

In [7]: s = input("Sentence : ")
        a = input("Search : ")
        b = input("Replace : ")

        if a in s:
            s = s.replace(a, b)
            print("Updated Sentence :", s)
        else:
            print("Word not found")

```

Updated Sentence : python is powerful to learn

21) Name Formatter

Problem Definition

Accept a person's name in any format and display:

- Name in Title Case
- Initials
- Surname
- Total number of characters (excluding spaces)

Sample

Input:
virat kohli

Output:
Formatted Name : Virat Kohli
Initials : V.K.
Surname : Kohli
Characters : 11

```
In [8]: n = input("Enter name: ")

n = " ".join(n.split())
p = n.split()

t = n.title()
i = ""

for x in p:
    i += x[0].upper() + "."

c = len(n.replace(" ", ""))

print("Formatted Name :", t)
print("Initials :", i)
print("Surname :", p[-1])
print("Characters :", c)
```

Formatted Name : Virat Kohli
Initials : V.K.
Surname : kohli
Characters : 10

22) Product Code Analyzer

Problem Definition

A product code is in the format ABC-1234-XY. Write a program to:

- Separate all three parts.
- Verify that: - first part contains only alphabets, - second part contains only digits, - third part contains only uppercase letters.

Sample

Input:
ABC-1234-XY

Output:

Code : ABC
Number : 1234
Category : XY
Valid Product Code

In [12]: `s = input("Enter code: ")`

`p = s.split("-")`

`if len(p) == 3:`

`a = p[0]`

`b = p[1]`

`c = p[2]`

`print("Code :", a)`

`print("Number :", b)`

`print("Category :", c)`

`if a.isalpha() and b.isdigit() and c.isupper() and c.isalpha():`
 `print("Valid Product Code")`

`else:`

`print("Invalid Product Code")`

`else:`

`print("Invalid Product Code")`

Code : aaa
Number : 2351
Category : X
Valid Product Code

In []: